Parallel beta-reduction contracts toward the complete development: if M bredpar M' then M' bredpar bcd(M)

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the case split on the four parallel-reduction rules, the two subcases inside the application rule and the final use of the substitution lemma are verbatim in the span; only the base variable rule is marked Exercise by the author, where M' = M and the claim is the reflexivity instance of the derivation rules.

Editorial note. The label lem:cont names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Editorial note. OLP reuses this label inside the unit; the card is identified by its statement span (source lines 857-859) and the claim key carries the statement start line.

Allowed standard prerequisites (editorial).

- Inductive definition of parallel beta-reduction. Same unit, rules \ollabel{defn:bredpar1}-\ollabel{defn:bredpar4} of Definition defn:bredpar. The four numbered cases of the derivation induction, each closed by the matching rule.
- Defining equations of the complete development bcd. Same unit, rules \ollabel{defn:bcd2}, \ollabel{defn:bcd3} of the bcd definition. Rewrite bcd(lambda x.N) and bcd(PQ) into the shapes the induction hypotheses fit.
- Substitution lemma for parallel reduction. Same unit, \ollabel{lem:comp} of the parallel-beta-reduction section (source line 811). Case 4: transports the pair of induction hypotheses through the substitution in a contracted redex.

Source reference. Original definition defn-bcd2: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 807–807. Prerequisite pointer invoked at proof line(s) 868; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-bcd3: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 809–809. Prerequisite pointer invoked at proof line(s) 882; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-bredpar1: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 770–770. Prerequisite pointer invoked at proof line(s) 863; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-bredpar2: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 771–771. Prerequisite pointer invoked at proof line(s) 864; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-bredpar3: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 773–773. Prerequisite pointer invoked at proof line(s) 870; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-bredpar4: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 775–775. Prerequisite pointer invoked at proof line(s) 879; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma lem-comp: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 825–825. Prerequisite pointer invoked at proof line(s) 888; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If $M \to_\beta M'$ then $M' \to_\beta \ulcorner M \urcorner$.*

Proof. By induction on the !!derivation of $M \to_\beta M'$.

1. The last rule is defn-bredpar1: Exercise.
2. The last rule is defn-bredpar2: M is $\lambda[x][N]$ and M' is $\lambda[x][N']$ with $N \to_\beta N'$. We want to show that $\lambda[x][N'] \to_\beta \ulcorner(\lambda[x][N])\urcorner$, i.e., $\lambda[x][N'] \to_\beta \lambda[x][\ulcorner N \urcorner]$ by defn-bcd2. It follows by defn-bredpar2 and the induction hypothesis.

3. The last rule is defn-bredpar3:M is PQ and M' is $P'Q'$ for some P, Q, P' and Q', with $P \to_\beta P'$ and $Q \to_\beta Q'$. By induction hypothesis, we have $P' \to_\beta \ulcorner P \urcorner$ and $Q' \to_\beta \ulcorner Q \urcorner$.

   (a) If P is $\lambda[x][N]$ for some x and N, then P' must be $\lambda[x][N']$ for some N' with $N \to_\beta N'$. By induction hypothesis we have $N' \to_\beta \ulcorner N \urcorner$ and $Q' \to_\beta \ulcorner Q \urcorner$. Then $(\lambda[x][N'])Q' \to_\beta \ulcorner N \urcorner[\ulcorner Q \urcorner / x]$ by defn-bredpar4.

   (b) If P is not a λ-abstract, then $P'Q' \to_\beta \ulcorner P \urcorner \ulcorner Q \urcorner$ by defn-bredpar3, and the right-hand side is $\ulcorner PQ \urcorner$ by defn-bcd3.

4. The last rule is defn-bredpar4: M is $(\lambda[x][N])Q$ and M' is $N'[Q'/x]$ for some x, N, Q, N', and Q', with $N \to_\beta N'$ and $Q \to_\beta Q'$. By induction hypothesis we know $N' \to_\beta \ulcorner N \urcorner$ and $Q' \to_\beta \ulcorner Q \urcorner$. By lem-comp we have $N'[Q'/x] \to_\beta \ulcorner N \urcorner[\ulcorner Q \urcorner / x]$, the right-hand side of which is exactly $\ulcorner((\lambda[x][N])Q)\urcorner$.

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